

Solitons and Energy Transfer Along Protein Molecules

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It is shown that the energy released in the hydrolysis of ATP molecules can be transferred in the form of vibration solitons along α -helical protein molecules. The vibration solitons are collective excitations travelling along a chain of successively arranged peptide groups and corresponding to amide I vibrations. The exceptional stability of solitons in one-dimensional structures can account for the small probability of their energy transforming into that of disordered heat motion.

1. Introduction

Many biological phenomena are known to be associated with spatial energy transfer along protein molecules. To elucidate the mechanism for this energy transfer at the molecular level is one of the most urgent problems in bioenergetics.

It is now established that the energy (about 0.43 eV or 10 kcal/mol) released in the hydrolysis of adenosine triphosphate molecules (ATP) is the universal energy unit used in all biological phenomena. This energy is only 20 times greater than the average energy of heat motion at physiologic temperatures. It is insufficient to excite the electronic states of molecules. Therefore the importance of vibration excitations (with an energy less than 0.4 eV) of separate groups of atoms in protein molecules has been repeatedly indicated.

The opponents of this viewpoint have frequently mentioned the fact that the lifetime (with respect to transformation into heat) of such excitations is very small (of the order of 10^{-12} – 10^{-13} s). At the same time it is usually forgotten that when the molecular system is organized in a certain manner the lifetime of excitations of some of its degrees of freedom can increase by many orders. For example, in condensed luminescent molecular systems (in particular, in crystalline anthracene) the electronic excitations of a system of conjugated π -electrons give off almost all their energy by radiation during the time (10^{-8} – 10^{-9} s) exceeding considerably the time of thermal relaxation.

In addition to the fact that the energy released in the hydrolysis of ATP molecules is insufficient to excite electronic states, the excitation of the latter is also disadvantageous because their lifetime relative to radiation is considerably smaller than the corresponding lifetime of vibration states. For example, the lifetime (relative to radiation) of valence vibrations $C=O$ (amide I with an energy of 0.21 eV) in peptide groups of proteins is equal to about 0.01 s. It exceeds the lifetime of electronic excitations of the same group (~ 4.4 eV of energy) by a factor of $(4.4/0.21)^4 = 1.9 \times 10^5$.

In alpha-helical protein molecules the peptide groups ($H-N-C=O$) are situated along the molecule in three chains. In each of these chains the neighbouring peptide groups are held together by hydrogen bonding between atomic H and O. Just these bonds provide for a relatively rigid structure of the α -helix.

Amide I vibrations are quite characteristic, i.e. they are weakly connected with other normal vibrations of protein molecules. This fact is confirmed by a very small difference in the amide I vibration frequencies in proteins of different kinds.

The collective excitations of a chain of successively arranged peptide groups which correspond to amide I vibrations are described by "solitary waves"—the vibration solitons. They have a particle-like character, propagate rapidly along the chain and have big lifespan, because they interact with other degrees of freedom of the molecule much more weakly than separate amide I vibrations.

The solitons were first observed as solitary, rapidly moving water loops in narrow channels filled with water. The "tsunami" covering huge distances in the oceans is also identified with solitons by some people. Numerous investigations (Hasegawa & Tappert, 1973; Scott, Chu & McLaughlin, 1973; Toda, 1975) showed that a remarkable property of the solitons described by the same non-linear equation as vibration solitons is their exceptional stability for outer disturbances.

The formation of stable excitations such as vibration solitons in the α -helical portions of protein molecules guarantees the high efficiency of the transfer of energy, released in the hydrolysis of ATP molecules, by protein. The exceptional stability of solitons in one-dimensional structures can account for the small probability of their energy transforming into that of disordered heat motion.

2. Solitons in One-Dimensional Chains

The dipole-dipole interaction between the adjacent peptide groups situated along a hydrogen-bonded chain in the α -helical protein molecules is an order of magnitude higher than a similar interaction between the

nearest groups of neighbouring chains. Therefore the excitations of the three chains may thus be considered independently. For the sake of simplicity we shall further investigate the excited states of only one of these chains.

Consider an infinite chain of weakly bound peptide group with mass M situated along axis x at distance R . Assume that the internal excitations of peptide groups are characterized by an energy ε (equal to 0.21 eV) and an electric dipole moment d directed along the chain. To investigate the collective excitations of this chain we shall use the results obtained by the author and Kislukha (1976).

Let x_n be a co-ordinate characterizing the position of the n th group and B^+ the creation operator for an amide I vibration excitation.

The excited state of such a chain is defined by the wave function

$$\psi(t) = \sum_n a_n(t) B_n^+ |0\rangle \quad (1)$$

with the normalized condition

$$\langle \psi(t) | \psi(t) \rangle = \sum_n |a_n(t)|^2 = 1. \quad (2)$$

Here $|0\rangle$ is the function of the ground state. The function (1) must satisfy the Schrodinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi.$$

Since the effective mass M of the peptide group is large, we treat their motion classically, and the amide I vibration excitations terms of quantum mechanics.

The Hamiltonian of the chain (provided the interaction between adjacent groups only is taken into account) can then be written as

$$H = T + U + \sum_n [(\varepsilon - D_n) B_n^+ B_n - J(B_{n+1}^+ B_n + B_{n+1} B_n^+)] \quad (3)$$

where the function $-D_n$ characterizes the change in the static interaction (the energy of the hydrogen bond) of the n th group with neighbouring groups when it is excited. Making allowance for the interaction with nearest neighbours we have

$$D_n = \mathcal{D}_n(|x_{n+1} - x_n|) + \mathcal{D}_n(|x_{n-1} - x_n|) \approx \left(1 + \frac{\beta}{R} \rho_n\right) D$$

where

$$\rho_n = R - |x_n - x_{n-1}|$$

is the decrease in equilibrium distances between peptide groups; $D = 2\mathcal{D}(R)$ and β are the positive parameters of a theory. $J = 2d^2/R^3$ is the operator

of the resonance interaction determining the transition of an excitation from one group to another;

$$T + U = \frac{1}{2} \sum_n \left[M \left(\frac{\partial x_n}{\partial t} \right)^2 + w \rho_n^2 \right]$$

are the kinetic and the potential energies of the displacement of peptide groups; w is the elasticity constant of the chain.

The set of functions $a_n(t)$ and $\rho_n(t)$ determine the distribution of excitations of peptide groups in the chain and the relative changes in the equilibrium distances between them. They obey the set of equations

$$i\hbar \frac{\partial a_n}{\partial t} = \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \rho_n \right) D \right] a_n - J(a_{n+1} - a_{n-1}), \quad (4)$$

$$\frac{\partial^2 \rho_n}{\partial t^2} = -\frac{w}{M} (2\rho_n - \rho_{n+1} - \rho_{n-1}) + \frac{\beta D}{MR} (2|a_n|^2 - |a_{n+1}|^2 - |a_{n-1}|^2). \quad (5)$$

Equations (4) and (5) define the collective excitations and deformations of the chain.

We are interested only in excitations distributed over some region of the chain. Therefore we can apply the continual model. At this approximation we can replace discrete functions $a_n(t)$ and $\rho_n(t)$ by the continuous functions $a(\xi, t)$ and $\rho(\xi, t)$ of the dimensionless co-ordinate $\xi = x/R$. Using the expression

$$f(\xi \pm 1, t) \approx \left(1 \pm \frac{\partial}{\partial \xi} + \frac{1}{2} \frac{\partial^2}{\partial \xi^2} \right) f(\xi, t),$$

we can transform equations (4) and (5) to

$$\left\{ i\hbar \frac{\partial}{\partial t} - \left[\lambda - \frac{\beta D}{R} \rho(\xi, t) \right] + J \frac{\partial^2}{\partial \xi^2} \right\} a(\xi, t) = 0, \quad (6)$$

$$\left[\frac{\partial^2}{\partial t^2} - v_a^2 \frac{\partial^2}{\partial \xi^2} \right] \rho(\xi, t) + \frac{\beta D}{MR} \frac{\partial^2}{\partial \xi^2} |a(\xi, t)|^2 = 0, \quad (7)$$

where

$$\begin{aligned} \lambda &\equiv \varepsilon + T + U - D - 2J, \\ T + U &\equiv \frac{M}{2} \int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t} \rho(\eta, t) \right)^2 + v_a^2 |a(\xi, t)|^4 \right], \end{aligned} \quad (8)$$

$v_a^2 \equiv w/M$, therefore $V_a = v_a R$ is the longitudinal sound velocity in the chain.

We shall look for solution of the set of equations (6) and (7) in the form of excitations propagating along the chain with velocity $V = vR$. For this

we set

$$\rho(\xi, t) = \rho(\xi - vt), \quad a(\xi, t) = \Phi(\xi - vt) \exp [i\theta(\xi, t)].$$

We then obtain from equation (7)

$$\rho(\xi, t) = \frac{\beta D |a(\xi, t)|^2}{MR(v_a^2 - v^2)}. \quad (9)$$

Substituting this value in equations (6) and (8) we transform them to

$$\left\{ i\hbar \frac{\partial}{\partial t} - \lambda + J \frac{\partial^2}{\partial \xi^2} + G_v |a(\xi, t)|^2 \right\} a(\xi, t) = 0, \quad (10)$$

$$T + U = \frac{G_v}{2(v_a^2 - v^2)} \int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t} |a(\eta, t)|^2 \right)^2 + v_a^2 |a(\xi, t)|^4 \right], \quad (11)$$

where

$$G_v = \frac{G_0}{1 - (V/V_a)^2}, \quad G_0 = \frac{\beta^2 D^2}{wR^2}. \quad (12)$$

The exact normalized solution of the non-linear Schrodinger equation (10) is called soliton. It has the form

$$a_v(\xi, t) = \frac{\sqrt{\mu_v/2} \exp \left\{ i \left[\frac{\hbar v}{2J} (\xi - \xi_0) - \Omega_v t \right] \right\}}{\cosh [\mu_v(\xi - \xi_0 - v)]}, \quad (13)$$

where

$$\mu_v = G_v/4J; \quad (14)$$

ξ_0 is the arbitrary constant due to the translational invariance of equations (6) and (7). Each value $V = vR$ corresponds to the soliton energy

$$E_V \equiv \hbar\Omega_V = \hbar\Omega_0 + \frac{\hbar^2 V^2}{2JR^2} + \frac{G_0^2(z^2 + \frac{3}{2}z^4 - \frac{1}{2}z^6)}{24J(1 - z^2)^3}, \quad (15)$$

where $z \equiv V/V_a$ and

$$\hbar\Omega_0 = \varepsilon - D - 2J - G_0^2/48J \quad (16)$$

is the internal energy of the soliton.

It follows from the expression (15) that when the soliton velocity approaches the longitudinal sound velocity V_a the soliton energy tends to infinity. At all finite values of the soliton energy its velocity is therefore always less than V_a . The soliton energy thus cannot transform into the energy of longitudinal sound waves. This is one of the reasons of the great stability of the solitons.

At small velocities of the soliton when $V \ll V_a$ is valid, the soliton energy (15) transforms to a simple expression

$$E_V = \hbar\Omega_0 + \frac{1}{2}m^*V^2, \quad (17)$$

where

$$m^* = \frac{1}{2J} \left[\frac{\hbar^2}{R^2} + \frac{G_0^2}{12V_a^2} \right] \quad (18)$$

is effective mass of the soliton. In the same approximation the parameter

$$\mu_v \approx \frac{G_0}{4J} [1 + (V/V_a)^2] \approx \frac{G_0}{4J} = \mu_0$$

is not dependent upon the soliton velocity.

Substituting equation (13) into equation (9) we find the distribution of the change in relative distances between the peptide groups. At small velocities of the soliton we shall get

$$\rho(\xi, t) = \frac{\sqrt{G_0^3/w}}{8J} \operatorname{sech}^2 [\mu_0(\xi - \xi_0 - vt)]. \quad (19)$$

According to equation (19), the maximal relative changes in distances between peptide groups at $\xi = \xi_0 + vt$ moves along the chain with the velocity $V = vR$. The deformation moving along the chain is accompanied by the internal excitation of the peptide groups. The region of the chain embraced by the excitation and deformation is defined by

$$R\Delta\xi \approx \frac{\pi R}{\mu} = \frac{4\pi JwR^3}{\beta^2 D^2}. \quad (20)$$

3. Excitons in One-Dimensional Chains

In an absolutely rigid chain $w = \infty$, so that the non-linearity parameter (12) is equal to zero and equation (10) reduces to linear Schrodinger equation which has solutions in the form of plane waves. In this case an excitation is uniformly distributed along the whole chain. The excitations of this type is called excitons. The internal energy of the exciton is

$$E_0 = \varepsilon - D - 2J. \quad (21)$$

The effective mass of the exciton is

$$m_0^* = \hbar^2/2R^2J. \quad (22)$$

At the final value w the non-linear equation (10) has also solution in the form

of plane wave (exciton)

$$a(\xi, t) = A \exp \{i[k\xi R - \omega(k)t]\} \quad (23)$$

which characterizes the uniform distribution of an exciton along the whole chain. To each wave (23) there corresponds the energy

$$\hbar\omega(k) = E_0 + \hbar^2 k^2 / 2m_0^*$$

The internal energy exciton E_0 and an effective mass m_0^* are determined by expressions (21) and (22). Therefore the internal energy of the exciton exceeds the internal energy (16) of the soliton by the value $G_0^2/48J$. Thus the excitations (23) in one-dimensional molecular systems are unstable.

In order to satisfy the normalization condition (2), we must set up in a chain of infinite length a wave packet with values of k in some narrow interval $k_0 - \Delta k \leq k \leq k_0 + \Delta k$, using the solution (23). The excitation described by such a wave packet propagates with the group velocity $V_g = \hbar k_0 / m_0^*$. In this state the energy and the momentum of the exciton have not well-defined values. The region of the chain embraced by the excitation increases, because these wave packets are known to "spread". If the excitation was initially localized on segment $(\Delta x)_0 = 2\pi/\Delta k$ this segment increases eventually, so that in the time t

$$(\Delta x)_t \approx \sqrt{(\Delta x)_0^2 + \hbar^2 t^2 / [m_0^*(\Delta x)_0]^2}.$$

4. The Stability of Solitons

A remarkable property of the vibration solitons at small velocities is that the distribution of excitation and deformation characterized by the function (19) remains completely unchanged in motion. The value of the maximum, the shape and the width (20) are independent of the velocity (energy) of a soliton and are defined only by the properties of the chain, i.e. the quantities J , G_0 , R . When the soliton slows down, its velocity changes, leaving the shape and the size of the excitation region unchanged. The stability property of solitons enables us to regard them as particle-like formations.

The stability of solitons is connected with the competition of two processes: the spreading of excitation region, determined by the resonance interaction J , and the tendency towards excitation localization which is due to the non-linear terms of equation (10) proportional to the parameter (12). In protein molecules the formation of solitons corresponding to the vibration amide I is favoured by the one-dimensional arrangement of peptide groups, the relative softness of hydrogen bonds between peptide groups (the small w) and the orientation of electric dipole moments of vibrations along the chain. Only if the electric dipole moments in the excitation region are arranged

in this manner the distances between peptide group decrease, thus promoting excitation localization. In very rigid chains the ratio G_0/J is very small and the formation of solitons is impossible.

Solitons can be formed only under a strong coupling of the internal vibration with a local deformation of the chain. This coupling is proportional to the non-linearity parameter $G_0 = \beta^2 D^2 / w R^2$. This parameter is big enough in comparatively soft chains of the peptide groups in protein molecules.

The condition of the formation solitons in α -helical protein molecules resembles the condition of the super conductivity formation in a metal. The superconductivity state due to the interaction of the electrons of a metal with phonons (the shifts of ions). Good conductors (silver, copper, gold) do not turn into a superconductive state. Superconductive states arise only in the metals which have a strong enough electron-phonon interaction. The strong electron-phonon interaction leading to a large resistivity in the normal state of metal makes the formation a superconducting state without any resistivity.

The above-considered solitons transfer with a very great efficiency the energy of excitation between different portions of the protein molecule. The energy transfer is accompanied by the deformation moving along the protein molecule. These solitons have a longitudinal polarization, since the electric dipole moment is directed along the chain.

The formation of solitons can occur only under the action of local excitations. These types of excitations also include the chemical energy released in the hydrolysis of ATP molecules which often attach themselves to certain portions of protein molecules (for example, to the heads of myosin molecules in the muscles). The vibration solitons cannot be excited by the infrared radiation of suitable frequency ($\sim 1660 \text{ cm}^{-1}$), since the wave length of this radiation is very large ($\sim 6 \times 10^4 \text{ \AA}$), and the photon localization is impossible with an accuracy higher than the wave length of radiation.

The concept of solitons which occur in myosin molecules was used by the present author[†] (Davydov, 1973, 1974, 1975) to elucidate the contraction mechanism of transversely striated muscles at the molecular level. Only the kinetic energy of solitons is actively used in muscle contraction, since the sliding of thin filaments relative to thick ones in the sarcomeres is due to the movement of solitons in myosin molecules contained in thick filaments.

[†] In these papers it was assumed that the distortion of the chain follows in an inert manner the movement of excitation. The notion of inert motion of peptide groups is valid only if the soliton velocity is much less than the longitudinal sound velocity in chain of the peptide groups.

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